COMP6841 PROJECT

Signal Forge CTF

A local training range for contested-spectrum security.

Radio signals feed software decisions. This project lets learners inspect that chain without transmit hardware.

Context

RF is **EXPENSIVE** and **easy** to break

Many people don't get to have these nice things…

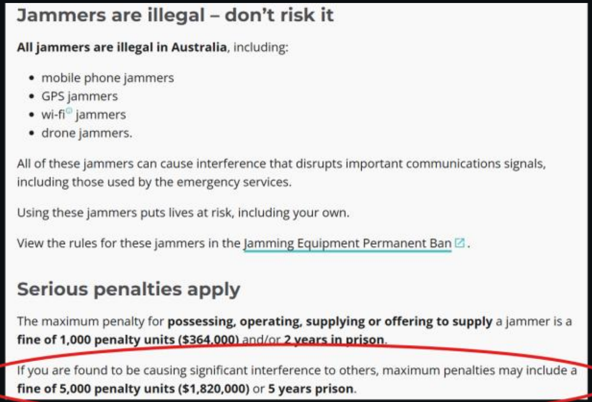
Transmitter And Processing

$3500 AUD

OR

$25000 AUD

Testing Environment



Jammers are illegal – don't risk it

All jammers are illegal in Australia, including:

- mobile phone jammers
- GPS jammers
- wi-fi jammers
- drone jammers.

All of these jammers can cause interference that disrupts important communications signals, including those used by the emergency services.

Using these jammers puts lives at risk, including your own.

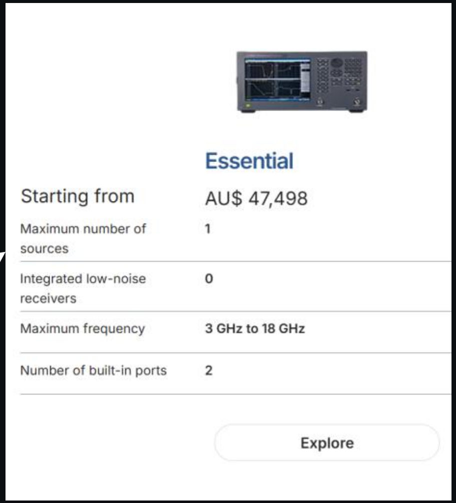
View the rules for these jammers in the Jamming Equipment Permanent Ban.

Serious penalties apply

The maximum penalty for **possessing, operating, supplying or offering to supply** a jammer is a **fine of 1,000 penalty units ($364,000)** and/or **2 years in prison**.

If you are found to be causing significant interference to others, maximum penalties may include a **fine of 5,000 penalty units ($1,820,000)** or **5 years prison**.

Receiving/Measurement



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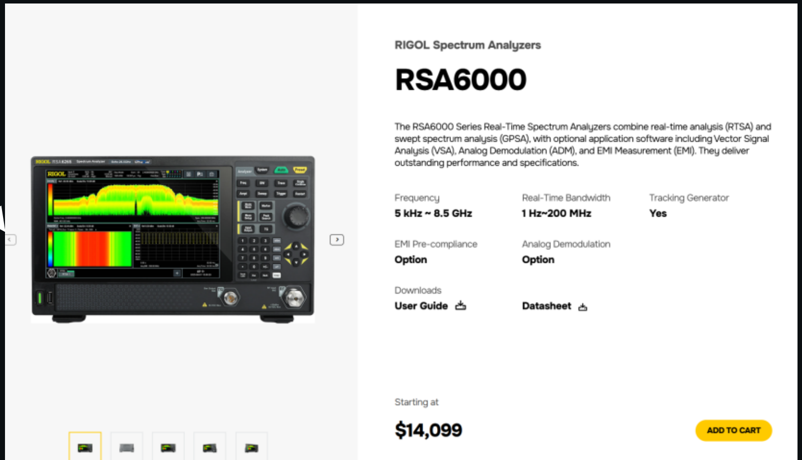
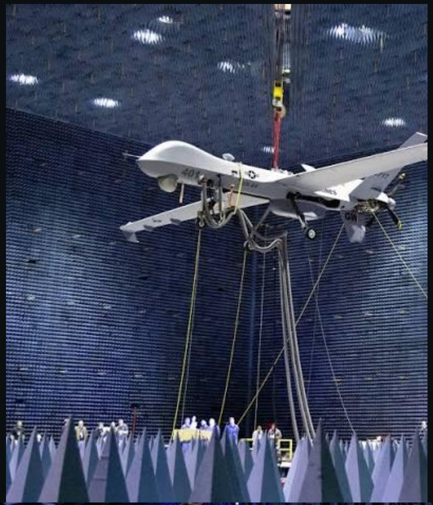
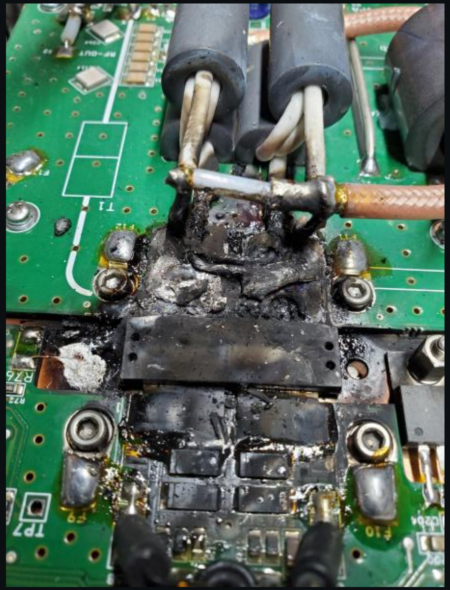
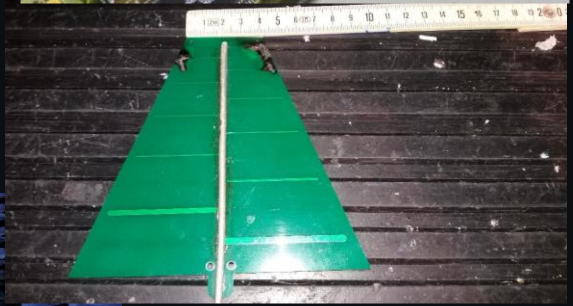


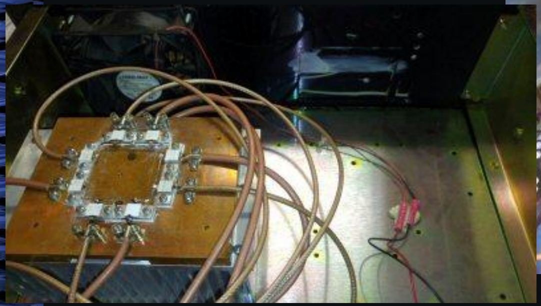
Context

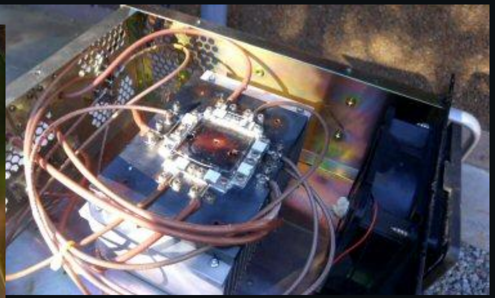
SOME People get these nice things… but is often given to the hardware people.





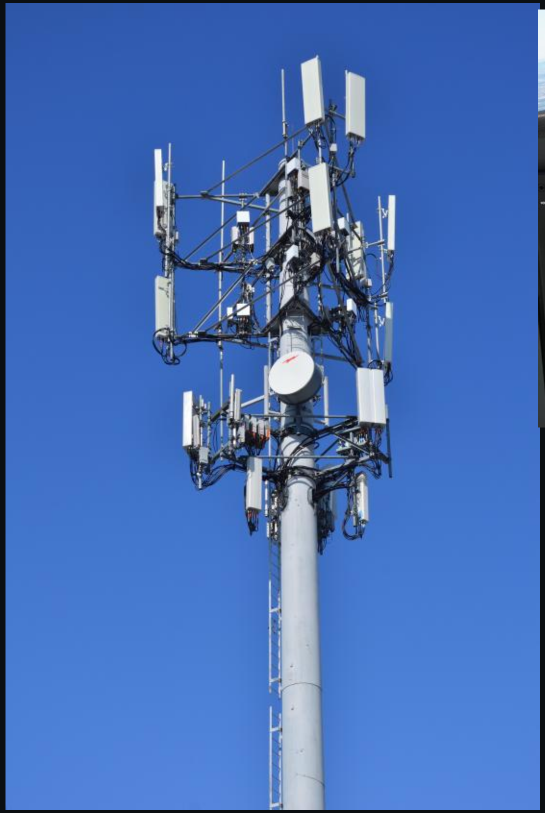






Context

When RF is used to control systems remotely, they mix data and control. Many critical systems depend on this architecture, but it also creates an attack surface: signals originate in hardware, while their interpretation and control logic are handled in software. But this may not be communicated as a vulnerability.



What is the threat model ?

The main ways an adversary can attack a radio link — and what each one targets.

ASSET — Anything That Uses Its Own RF Link For Control
The over-the-air link that carries telemetry and control between hardware and software.

VULNERABILITIES

01
DoS
Flooding the channel with noise or jamming to drown out the real signal and deny service to legitimate receivers.

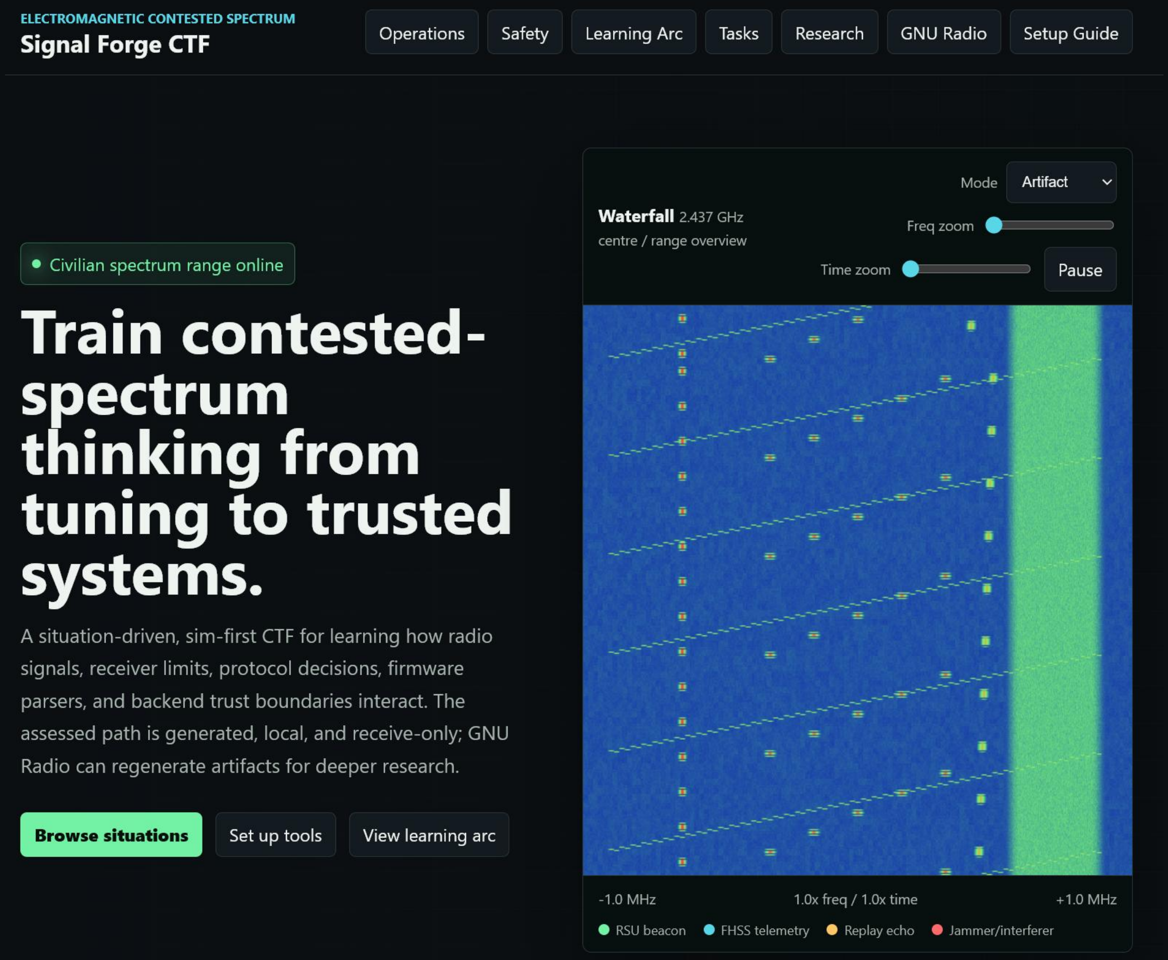
02
Spoofing
Transmitting forged or replayed signals that impersonate a trusted source to deceive the receiver.

03
Intelligence gathering
Passively intercepting and analysing transmissions to map activity and extract sensitive information.

SIGNAL FORGE CTF

Signal Forge

It is a structured lab for RF analysis, cyber exploitation, and defensive reasoning.



Current contexts cover the beginner-to-advanced ladder

The structure is already in place; the next pass is making each signal scheme distinct and more realistic.

Context	Level	Subtasks	Focus
Tunnel Infrastructure	Basic	4	signals + cema
Song Metadata	Basic	4	signals + cema + cyber
Weather Broadcast	Moderate	3	signals + cyber
Civilian Emergency Radio Audio Network	Moderate	3	signals + cyber
~~Bushfire C2~~	Advanced	4	cyber + cema
~~Farm Gates~~	Advanced	4	cyber + cema

Security Concepts

Multiple contexts available — the security task types implemented across the range.

01
Jamming
Deliberately transmitting interference to disrupt or deny a target signal.

02
Anti-jamming measures
Keeping a link usable under interference with spread-spectrum and filtering techniques.

03
Coding
Data coding — encoding schemes that structure and protect data sent over the air.

04
Scrambling
Randomising the bitstream to whiten the spectrum and obscure structure.

05
Reverse Engineering
Recovering framing, symbols and protocol fields from an unknown signal.

06
Signal Processing
Filtering communications — conditioning received signals to isolate the channel of interest.

Tangible Examples

- These are the same signal, one has prng phase swaps the other does
- But different views differ dramatically but is it secure!
- Maybe there is an insecurity if we consider groupings